

If the lightest neutrino mass is the dark-energy scale: a narrow window for neutrinoless double-beta decay

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The observed dark-energy density corresponds to an energy scale $\rho_\Lambda^{1/4} \simeq 2.25 \text{ meV}$, close to the neutrino mass splittings. This numerical proximity is long noted and has motivated a substantial model-building literature. We do not propose a mechanism. Instead we take the sharpest arithmetic version of the coincidence as a hypothesis — that the lightest neutrino mass equals that scale, $m_1 = 2.25 \text{ meV}$, with normal ordering — and ask what it implies for neutrinoless double-beta decay. The mapping from m_{lightest} to the effective Majorana mass is standard and the resulting band appears in every review of the subject; what is specific here is the consequence of evaluating it at this one value. With current oscillation parameters the three contributions are $(1.52, 2.67, 1.10) \text{ meV}$, giving $m_{\beta\beta} \in [0.04, 5.30] \text{ meV}$ and $\Sigma m_\nu = 61.3 \text{ meV}$. The lower edge is not protected by any symmetry: it is nonzero only because the middle term exceeds the other two combined by 0.045 meV , so the three phasors cannot close. That same near-cancellation makes $m_{\beta\beta}$ an unusually steep function of m_1 . The observationally useful statement is the upper edge. Minimal normal ordering ($m_1 = 0$) cannot exceed 3.69 meV at any phases, while this hypothesis reaches 5.30 meV and occupies the interval between them 32% of the time. The projected reach of a single planned experiment falls inside that interval, so a detection there would be difficult to attribute to minimal ordering, while a detection above 5.30 meV excludes the hypothesis outright. A null result constrains nothing, at any sensitivity.

I. INTRODUCTION

The energy density of dark energy, expressed as a mass scale, is

$$\rho_\Lambda^{1/4} \simeq 2.25 \text{ meV}, \quad (1)$$

which lies within an order of magnitude of $\sqrt{\Delta m_{21}^2} \simeq 8.6 \text{ meV}$ and two orders of $\sqrt{\Delta m_{31}^2} \simeq 50 \text{ meV}$. That the cosmological constant and the neutrino masses should sit at comparable scales, when they are separated by every other consideration in the Standard Model, has been remarked on for decades and has motivated a large body of model building — most directly the mass-varying-neutrino scenarios in which the dark-energy density is tied dynamically to the neutrino sector [1, 2].

We take no position on any of those mechanisms, and we propose no new one. Our purpose is narrower and, we think, more useful: the coincidence in Eq. (1) admits one arithmetically sharp reading, and that reading has consequences which are close to being tested. The reading is

$$m_1 = \rho_\Lambda^{1/4} = 2.25 \text{ meV}, \quad \text{normal ordering}, \quad (2)$$

with m_1 the lightest neutrino mass. Everything below follows from Eq. (2) together with measured oscillation parameters. No cosmological model enters, and nothing we compute depends on why Eq. (2) might hold.

What is and is not new here. The map from the lightest mass to the effective Majorana mass $m_{\beta\beta}$ is textbook,

and the familiar lobster-shaped allowed region in the $(m_{\text{lightest}}, m_{\beta\beta})$ plane appears in every review of neutrinoless double-beta decay [3, 4]. Evaluating it at one particular value of m_{lightest} is not a new calculation. What we draw out is that this particular value lands in an unusually informative place: it sits just above the reach of minimal normal ordering, and the gap between the two is where one planned experiment is projected to look. That placement, and the detection probabilities attached to it, is the content of this paper.

Section II fixes the inputs and computes the window. Section III explains why the lower edge is nonzero and why it is fragile. Section IV sets out what would distinguish the hypothesis from minimal ordering and what would refute it. Section V states the assumptions.

II. THE EFFECTIVE MASS

For three Majorana neutrinos the amplitude for neutrinoless double-beta decay is controlled by

$$m_{\beta\beta} = \left| \sum_{i=1}^3 U_{ei}^2 m_i \right| \quad (3)$$

$$= \left| c_{12}^2 c_{13}^2 m_1 + s_{12}^2 c_{13}^2 m_2 e^{i\alpha_2} + s_{13}^2 m_3 e^{i\alpha_3} \right|,$$

with α_2, α_3 the Majorana phases, which oscillation experiments do not constrain. We use $\sin^2 \theta_{12} = 0.307$, $\sin^2 \theta_{13} = 0.022$, $\Delta m_{21}^2 = 7.42 \times 10^{-5} \text{ eV}^2$ and $\Delta m_{31}^2 = 2.51 \times 10^{-3} \text{ eV}^2$, consistent with current global fits [5, 6]. With $m_2 = (m_1^2 + \Delta m_{21}^2)^{1/2}$ and $m_3 = (m_1^2 + \Delta m_{31}^2)^{1/2}$, Eq. (2) gives $m_2 = 8.90 \text{ meV}$, $m_3 = 50.15 \text{ meV}$, and

$$\Sigma m_\nu = 61.3 \text{ meV}. \quad (4)$$

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The three contributions to Eq. (3) are

$$(|U_{e1}^2 m_1|, |U_{e2}^2 m_2|, |U_{e3}^2 m_3|) = (1.52, 2.67, 1.10) \text{ meV}. \quad (5)$$

Varying the phases over their full range gives

$$0.04 \text{ meV} \leq m_{\beta\beta} \leq 5.30 \text{ meV} \quad (6)$$

with $\simeq 3.3 \text{ meV}$ typical. The hypothesis fixes m_1 but says nothing about the Majorana phases, so position within Eq. (6) is unconstrained; that is a genuine limitation and not a marginal one, as Sec. IV makes quantitative.

The mass sum in Eq. (4) is the more immediately testable number. Current cosmological bounds from the microwave background combined with baryon acoustic oscillations have reached $\Sigma m_\nu \lesssim 72 \text{ meV}$, with some combinations pressing lower [7]. The value in Eq. (4) sits just inside, so this hypothesis may well be graded by cosmology before any laboratory experiment reaches it.

III. WHY THE LOWER EDGE IS NONZERO, AND WHY IT IS FRAGILE

Complete cancellation in Eq. (3) requires the three complex terms to close a triangle, which is possible only if no one of them exceeds the sum of the other two. From Eq. (5),

$$2.67 > 1.52 + 1.10 = 2.62 \text{ meV}, \quad (7)$$

by 0.045 meV . The middle term wins, the triangle cannot close, and the lower edge of Eq. (6) follows as the residue $2.67 - 2.62$.

Two things should be said about this immediately. First, the margin is 0.045 meV on terms of order 2.6 meV — a 1.7% effect. Nothing protects it. It is not the consequence of a symmetry and it would not survive a modest shift in θ_{12} , in Δm_{21}^2 , or in m_1 itself. At $m_1 \gtrsim 2.32 \text{ meV}$ the inequality reverses and the floor vanishes entirely.

Second, and more usefully, the same near-degeneracy makes $m_{\beta\beta}$ a steep function of m_1 near this point. A quantity sitting close to a cancellation responds sharply to its inputs. This is the one respect in which evaluating the standard band at this particular value is more than a lookup: the hypothesis places m_1 where the effective mass is unusually sensitive to it.

We stress that no observable consequence rides on the lower edge. At 0.04 meV it is two orders of magnitude below any experimental reach on any timeline under discussion. Every statement in Sec. IV concerns the upper edge, which the near-cancellation barely moves.

IV. WHAT WOULD TEST IT

A. The interval that separates hypotheses

The comparison that matters is not against zero but against the minimal alternative. Setting $m_1 = 0$ with the

same ordering and parameters gives

$$m_{\beta\beta}^{\min} \in [1.48, 3.69] \text{ meV}, \quad (8)$$

so minimal normal ordering has a hard ceiling at 3.69 meV that no choice of phases can exceed. The hypothesis of Eq. (2) reaches 5.30 meV . The interval

$$3.69 \text{ meV} < m_{\beta\beta} < 5.30 \text{ meV} \quad (9)$$

is therefore inaccessible to minimal ordering and occupied by this hypothesis. Integrating Eq. (3) over flat Majorana phases, the hypothesis lands inside Eq. (9) 31.8% of the time.

B. Where the experiments fall

Projected ten-year reaches for the ton-scale programme, quoted as ranges because the nuclear matrix element is the dominant uncertainty, are approximately 4.7–20.3 meV for nEXO [8], 9–21 meV for LEGEND-1000 [9], and 12–34 meV for CUPID [10].

Against the ceiling of Eq. (6), LEGEND-1000 and CUPID lie entirely above the hypothesis and cannot reach it for any matrix element. Only nEXO overlaps, and only at the favourable end of its matrix-element range, 4.7–5.3 meV. Over flat phases the hypothesis exceeds 4.7 meV 10.8% of the time.

Two features of this placement are worth separating. The probability is low — roughly one in nine — so the baseline experiment is unlikely to see a signal even if the hypothesis is correct. But the entire 4.7–5.3 meV overlap lies inside the interval of Eq. (9), which minimal ordering cannot reach. A baseline nEXO detection would therefore be hard to accommodate within minimal normal ordering, which is a stronger statement than the detection probability alone suggests.

C. Why a sensitivity upgrade does not help as much as expected

A factor-of-four improvement in half-life sensitivity, of the kind a barium-tagging upgrade is projected to deliver, corresponds to a factor $\sqrt{4} = 2$ in $m_{\beta\beta}$, taking the reach to $\simeq 2.35 \text{ meV}$. The detection probability rises from 10.8% to 69.1%.

This buys likelihood, not discrimination. Minimal normal ordering also exceeds 2.35 meV , and does so 63.7% of the time. A tagged detection near that threshold would be consistent with both hypotheses and would separate almost nothing. The unimproved machine is the discriminating one, precisely because its reach sits in the interval where the two hypotheses do not overlap.

D. Falsification

A measured $m_{\beta\beta} > 5.30$ meV refutes Eq. (2) outright, since no choice of phases can produce it. Evidence for inverted ordering does the same: at $m_3 = 2.25$ meV inverted ordering gives $m_{\beta\beta} \in [18.6, 49.1]$ meV, which does not overlap Eq. (6) at any phase. Evidence that neutrinos are Dirac particles removes the observable entirely.

A null result refutes nothing. Because the phases are free, $m_{\beta\beta}$ can be arbitrarily small, and no non-observation at any sensitivity constrains Eq. (2). This asymmetry is intrinsic to the hypothesis and we make no attempt to dress it up.

For orientation, the strongest current limit comes from ^{136}Xe : KamLAND-Zen reports $T_{1/2} > 4.3 \times 10^{26}$ yr at 90% confidence over a 2097 kg yr exposure, corresponding to $m_{\beta\beta} < 28\text{--}122$ meV across the matrix-element range [11]. That is a factor of five or more above the ceiling of Eq. (6), so no existing measurement bears on this hypothesis in either direction.

V. WHAT IS ASSUMED

Majorana neutrinos. Eq. (3) presumes lepton-number violation. If neutrinos are Dirac there is no $m_{\beta\beta}$ and the question does not arise.

Normal ordering. Assumed throughout, and stated in Eq. (2) rather than derived. Global fits currently prefer it at modest significance; the preference is not settled.

Three light species. A light sterile state would add a term to Eq. (3) and widen the window.

Standard mechanism. We assume light-neutrino exchange dominates the amplitude. If other lepton-number-violating physics contributes, $m_{\beta\beta}$ as extracted

from a half-life is no longer the quantity computed here.

The numerical input. We use the observed $\rho_\Lambda^{1/4} = 2.25$ meV. Nothing depends on this at the quoted precision: shifting it by 0.5% moves the lower edge of Eq. (6) from 0.045 to 0.039 meV and leaves the upper edge at 5.30 meV, so every conclusion in Sec. IV is unchanged.

And the coincidence itself. Eq. (2) is a hypothesis, not a result. We are aware that equating two scales because they are numerically close is weak grounds, and we do not argue otherwise. The case for writing it down is that it is sharp, that it costs nothing to state, and that it happens to be nearly testable — not that the coincidence is evidence for itself.

VI. CONCLUSION

Taking the lightest neutrino mass equal to the dark-energy scale, with normal ordering, gives $m_{\beta\beta} \in [0.04, 5.30]$ meV and $\Sigma m_\nu = 61.3$ meV. The lower edge exists only through a 0.045 meV margin in the triangle inequality and carries no observational weight. The upper edge does the work.

The useful comparison is against minimal normal ordering, whose hard ceiling is 3.69 meV. The interval between the two ceilings is closed to minimal ordering, is occupied by this hypothesis about a third of the time, and contains the entire projected overlap of one planned experiment. A detection there would be awkward to explain with minimal ordering; a detection above 5.30 meV would refute the hypothesis; a null result would say nothing at all.

Cosmology will likely speak first. A mass sum of 61.3 meV sits just inside current combined limits, and the bound is tightening.

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